DRAFT

Research Proposal

Evaluating the Integrated Open-Access DAEDALUS Explore Simulation and Code-Along Pedagogy in Health Systems Leadership

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Background & Rationale

There exists a documented deficiency in health administration education: students are frequently instructed in advanced health systems theories or discrete quantitative techniques; however, they are seldom trained to integrate both in high-stakes executive settings. Contemporary population health management necessitates a workforce proficient in core competencies, particularly data literacy and advanced analytics.[1] Although internet-based simulation systems have traditionally been employed in healthcare management training to facilitate the transfer of theoretical knowledge into practical decision-making skills, a significant gap persists in instructing executives on how to independently scrutinize the underlying computational models that underpin these simulations.[2,3]

Increasingly, health policy makers rely on computer disease simulation models to predict long-term outcomes and guide resource utilization strategies.[4,5] However, treating these models as impenetrable "black boxes" leaves non-technical administrators vulnerable to misinterpreting critical macroeconomic and epidemiological trade-offs. Furthermore, a primary barrier to expanding simulation-based and analytical education in health systems is the prohibitive economics of delivery. Simulation-based medical education requires substantial resource investment, and proprietary statistical software (e.g., `R`, `SAS`, `Stata`) imposes high licensing costs, thereby restricting widespread curricular integration.[6]

The recently redeveloped MHIA 6160 curriculum, part of the Masters of Healthcare Industry Administration program at the Schulich School of Business, York University, aims to address this evolving industry need through an innovative, highly scalable pedagogical framework. This framework overcomes traditional financial and temporal development barriers by utilizing exclusively open-source infrastructure: a *Code-Along* approach employing free R and `RStudio` software, implemented in conjunction with a capstone simulation utilizing the publicly accessible DAEDALUS Explore epidemiological and macroeconomic modelling interface.[7]

By removing the economic barriers to high-fidelity simulation and data analytics, the curriculum specifically addresses the remaining cognitive barriers: technical anxiety and coding self-efficacy. By converting DAEDALUS Explore outputs into strategic insights through reproducible `R` scripts, the course equips non-programmers with the essential data literacy necessary to effectively manage complex public health mandates and private-sector economic challenges.

Research Aim & Objectives

The primary objective of this prospective study is to assess the educational impact of an open-source, dual-intervention curriculum on the data literacy and self-efficacy of students in health systems management. By eliminating the conventional financial obstacles associated with simulation and statistical software, the study aims to isolate and investigate the cognitive and pedagogical barriers to executive technical proficiency.

Specifically, the study will investigate and quantify:

- The longitudinal decrease in technical anxiety and the corresponding enhancement in coding self-efficacy over a period of six weeks.
- The perceived utility of standardized *Code-Along* scaffolding in transforming raw, open-source programming exercises into tangible executive insights and visualizations.
- The effectiveness of the DAEDALUS Explore simulation in proficiently educating non-technical leaders to successfully interrogate computational "black boxes" and to operationalize macroeconomic health trade-offs (i.e., value versus quality of healthcare).
- Student perceptions concerning the accessibility and scalability of utilizing entirely open-source tools (`R`, `RStudio`, DAEDALUS Explore) for advanced health systems education.

Study Design & Methodology

We propose a prospective, mixed-methods longitudinal study design embedded directly within the standard course evaluations. To rigorously isolate actual competence from perceived confidence, the evaluation features both self-reported metrics and an ungraded, isomorphic strategic interpretation task.

- **Study Population:** Graduate students enrolled in MHIA 6160 (Winter 2027).
- **Data Collection:** Anonymous, integrated surveys and tasks administered across three milestones:

Wave	Milestone	Evaluation Focus
Wave 1	Week 1	**Baseline:** Assessment of technical anxiety and coding self-efficacy, paired with an isomorphic pre-test (evaluating a static DAEDALUS output) to establish baseline data interpretation skills.
Wave 2	Week 4	**Mid-point:** Evaluation of the *Code-Along* framework's impact on reducing cognitive load and building technical confidence.
Wave 3	Week 6	**Final:** Assessment of simulation integration and overall executive data literacy, paired with an isomorphic post-test requiring students to autonomously generate and interpret a strategically equivalent DAEDALUS output.

- **Quantitative Analysis:** Likert-scale responses will be analyzed using paired statistical testing to track longitudinal shifts in coding self-efficacy, technical anxiety, and the perceived accessibility of the open-source toolset (`R`, `RStudio`, and DAEDALUS Explore).

- **Qualitative & Performance Analysis:** Open-ended survey responses will undergo an inductive thematic analysis. Furthermore, the isomorphic pre- and post-tests will be evaluated using a standardized rubric to measure objective improvements in strategic decision-making, specifically capturing the transition from technical novices to managers capable of independently interrogating the computational "black box."

Stakeholder Value Proposition

- **Schulich School of Business:** By pioneering this scalable, open-source pedagogical model, the health industry management program exemplifies an evidence-based dedication to curriculum innovation. Disseminating these findings will underscore the program's leadership in addressing a vital industry competency gap, preparing graduates to adeptly navigate complex resource allocation and health policy decisions without the friction of prohibitive software costs.
- **Imperial College London:** This research introduces a novel application that enhances the DAEDALUS Explore interface, transforming it from a leading global epidemiological modelling tool into a fundamental educational resource. It showcases DAEDALUS Explore's distinctive ability to prepare future health system executives to analyze macroeconomic trade-offs, thereby extending its influence into postgraduate management pedagogy.

Actions & Timeline

- **Summer 2026:** Finalize curriculum architecture, open-source R-Laboratory materials, and the digital *Code-Along* staging area.
- **Fall 2026:** Submit the study protocol to the Research Ethics Board (REB) for expedited review and approval.
- **Winter 2027:** Execute the mixed-methods data collection across the three designated waves during the inaugural delivery of MHIA 6160.
- **Spring 2027:** Conduct quantitative and qualitative thematic data analysis, and draft the manuscript for a high-impact pedagogical or health administration journal.
- **Summer 2027:** Facilitate cross-institutional co-author review and formal manuscript submission.

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